

ASSESSMENT OF PRECISION WITH AVERSITY TO OVERSTATEMENT

Nicholas T. Longford

SNTL and Universitat Pompeu Fabra, Barcelona, Spain

e-mail: *sntl@nick@sntl.co.uk*

Key words: χ^2 distribution; aversity; LINEX loss; loss function; sampling variance.

Summary: The sampling variance and standard error of an estimator are usually estimated with small or no bias. An argument is put forward that they should be overestimated, to reflect our aversity to overstating the precision (understating the variance or standard deviation). Such aversity is characterized by an asymmetric loss function. The minimum-expected-loss estimators of the sampling variances and standard errors of parameters in ordinary regression are derived for several classes of loss functions: piecewise quadratic, linear, multiplicative and reciprocal, and LINEX.

1. Introduction

Reporting the results of estimation by the estimate, $\hat{\theta}$, and its estimated standard error, $\hat{\sigma} = \sqrt{\widehat{\text{var}}(\hat{\theta})}$, is a well established convention. In the ideal adopted by many, the estimator $\hat{\theta}$ is unbiased and efficient (has minimum mean-squared error), and its sampling variance σ^2 is estimated without bias. The estimate is the principal element of the output. We interpret the estimated standard error, or the estimated sampling variance, as a claim made by the analyst, akin to a warranty associated with a product purchased in a store. If the product turns out to outlive its warranty, the purchaser is not liable for the extra value received, although the retailer and manufacturer could have made a stronger claim (e.g., in marketing), and could have charged a higher price to reflect it. If the product fails earlier than claimed by the warranty the purchaser has a good case for compensation, and the retailer and producer incur other losses (costs), including loss of good reputation and of future custom.

Motivated by this example, we regard understatement, that is, $\hat{\sigma}^2 < \sigma^2$, as a more serious error than overstatement, $\hat{\sigma}^2 > \sigma^2$, by the same absolute error $\Delta = |\hat{\sigma}^2 - \sigma^2|$ or another measure of the size of the error; we are *averse* to understatement of σ^2 . Such aversity can be incorporated in the estimation of σ^2 by using an asymmetric loss function instead minimising the bias or the conventional quadratic loss. We derive the estimators of σ and σ^2 that have minimum expected loss with piecewise linear, quadratic, reciprocal and multiplicative loss functions. We argue that such estimators enhance the integrity of the statistical analysis by tailoring it to the client's perspective and purpose. For another application with these loss functions, see Longford (2010). We also study

estimators with minimum expected loss with respect to the LINEX loss function introduced by Varian (1975) for a specific problem and explored in detail by Zellner (1986). With LINEX, the loss is related to the absolute error Δ approximately linearly for one sign of the error and approximately exponentially for the other sign. For general background to analysis with loss functions (utilities), we refer to DeGroot (1970), Berger (1985) and Lindley (1985).

The next section formalizes the problem of estimating with aversity to understatement and derives estimators of the sampling variance σ^2 and standard error σ of a parameter in ordinary regression. Variance estimation with the LINEX loss function is explored in Section 3. The concluding section summarizes the results and presents arguments for adopting our perspective.

2. Estimating with minimum expected loss

Suppose $\hat{\sigma}^2$ has the scaled χ^2 distribution with k degrees of freedom; that is, $k\hat{\sigma}^2/\sigma^2$ has the distribution with density

$$f_k(x) = \frac{1}{2\Gamma(\frac{1}{2}k)} \left(\frac{x}{2}\right)^{k/2-1} \exp\left(-\frac{x}{2}\right).$$

Let F_k be the corresponding distribution function. Suppose the loss is $\hat{\sigma}^2 - \sigma^2$ when $\hat{\sigma}^2 > \sigma^2$, and $R(\sigma^2 - \hat{\sigma}^2)$ when $\hat{\sigma}^2 < \sigma^2$; this loss function is piecewise linear. We refer to the positive scalar R as the (coefficient of) *aversity* to understatement. Of interest are values $R > 1$, although $R \in (0, 1)$ can also be considered. We seek the estimator $c\hat{\sigma}^2$ for which the expected loss, equal to

$$L = R \int_0^{\sigma^2} (\sigma^2 - x) \frac{k}{c\sigma^2} f_k\left(\frac{kx}{c\sigma^2}\right) dx + \int_{\sigma^2}^{+\infty} (x - \sigma^2) \frac{k}{c\sigma^2} f_k\left(\frac{kx}{c\sigma^2}\right) dx,$$

is minimized. By applying the transformation $z = kx/(c\sigma^2)$ and the identity

$$xf_k(x) = kf_{k+2}(x), \quad (1)$$

we obtain the equivalent expression

$$L = \sigma^2 \left[(R+1) \left\{ F_k\left(\frac{k}{c}\right) - cF_{k+2}\left(\frac{k}{c}\right) \right\} + c - 1 \right]. \quad (2)$$

In its derivative with respect to c ,

$$\frac{\partial L}{\partial c} = \sigma^2 \left[-\frac{k}{c^2} (R+1) \left\{ f_k\left(\frac{k}{c}\right) - cf_{k+2}\left(\frac{k}{c}\right) \right\} - (R+1)F_{k+2}\left(\frac{k}{c}\right) + 1 \right],$$

the expression in the braces vanishes owing to (1), and so it has the (unique) root

$$c_R^*(k) = \frac{k}{F_{k+2}^{-1}\left(\frac{1}{R+1}\right)}, \quad (3)$$

at which the expected loss attains its minimum, equal to

$$\sigma^2 \left[(R+1)F_k \left\{ F_{k+2}^{-1}\left(\frac{1}{R+1}\right) \right\} - 1 \right].$$

The expected loss of the unbiased estimator $\hat{\sigma}^2$ is $\sigma^2(R+1)\{F_k(k) - F_{k+2}(k)\}$.

For estimating σ by $d\hat{\sigma}$, where d is a scalar to be found, the expected loss is

$$\begin{aligned} L &= R \int_0^\sigma (\sigma - x) \frac{2kx}{d^2\sigma^2} f_k\left(\frac{kx^2}{d^2\sigma^2}\right) dx + \int_\sigma^{+\infty} (x - \sigma) \frac{2kx}{d^2\sigma^2} f_k\left(\frac{kx^2}{d^2\sigma^2}\right) dx \\ &= \sigma \left[(R+1) \left\{ F_k\left(\frac{k}{d^2}\right) - \frac{d\gamma_k}{\sqrt{k}} F_{k+1}\left(\frac{k}{d^2}\right) \right\} + \frac{d\gamma_k}{\sqrt{k}} - 1 \right], \end{aligned}$$

where $\gamma_k = \sqrt{2}\Gamma(\frac{1}{2}k + \frac{1}{2})/\Gamma(\frac{1}{2}k)$. In the derivation, we used the identity

$$\sqrt{x}f_k(x) = \gamma_k f_{k+1}(x).$$

The derivative of the expected loss with respect to d ,

$$\frac{\partial L}{\partial d} = \frac{\gamma_k \sigma}{\sqrt{k}} \left\{ 1 - (R+1) F_{k+1}\left(\frac{k}{d^2}\right) \right\},$$

has the unique root

$$d_R^*(k) = \frac{\sqrt{k}}{\sqrt{F_{k+1}^{-1}\left(\frac{1}{R+1}\right)}}.$$

The functions $c_R^*(k)$ and $d_R^*(k)$ are drawn in Figure 1 for a selection of aversity coefficients R and $5 \leq k \leq 60$. The functions converge to unity as $k \rightarrow +\infty$, but do so very slowly. For example, for $2 \leq R \leq 50$, $c_R^*(60)$ attains values in the range $(1.06, 1.45)$, and $c_R^*(1000) \in (1.02, 1.10)$. Note that $d_R^*(k)$ differs from $\sqrt{c_R^*(k)}$ only in the degrees of freedom used with the quantile function F_h^{-1} , $h = k+1$ vs. $h = k+2$. In general, $d_R^*(k) > \sqrt{c_R^*(k)}$, but the difference is appreciable only for small k . For $R = 1$, $c_1^*(k)$ and $d_1^*(k)$ are smaller than unity.

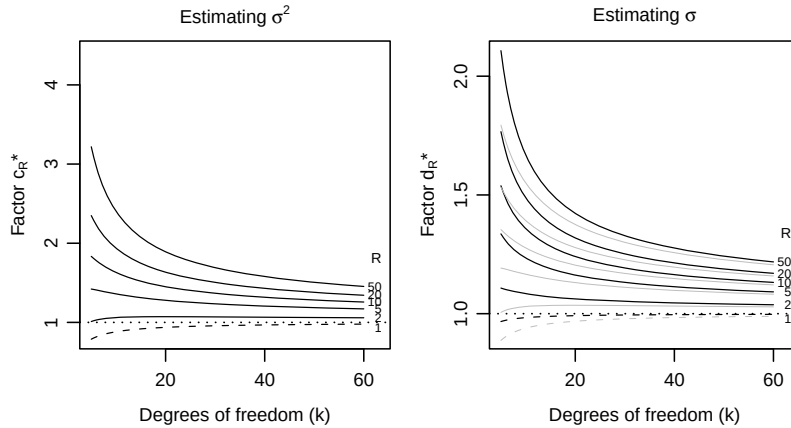


Figure 1: The factors c_R^* and d_R^* for which $c_R^*\hat{\sigma}^2$ and $d_R^*\hat{\sigma}$ are associated with minimum expected piecewise linear loss, as functions of the degrees of freedom $5 \leq k \leq 60$ and aversity R indicated at the right-hand margins. In the right-hand panel, the functions $\sqrt{c_R^*}$ are drawn by gray colour.

Estimation with the minimum mean-squared error corresponds to the quadratic loss function, $(\hat{\sigma}^2 - \sigma^2)^2$ or $(\hat{\sigma} - \sigma)^2$, which is subsumed in the class of piecewise quadratic functions; when $\hat{\sigma}^2 < \sigma^2$ the loss is equal to $R(\hat{\sigma}^2 - \sigma^2)^2$, and when $\hat{\sigma}^2 > \sigma^2$ the loss is $(\hat{\sigma}^2 - \sigma^2)^2$. The expected loss in estimating σ^2 by $c\hat{\sigma}^2$ is

$$\begin{aligned} L &= R \int_0^{\sigma^2} (x - \sigma^2)^2 \frac{k}{c\sigma^2} f_k\left(\frac{kx}{c\sigma^2}\right) dx + \int_{\sigma^2}^{+\infty} (x - \sigma^2)^2 \frac{k}{c\sigma^2} f_k\left(\frac{kx}{c\sigma^2}\right) dx \\ &= \sigma^4(R-1) \left\{ c^2 \frac{k+2}{k} F_{k+4}\left(\frac{k}{c}\right) - 2c F_{k+2}\left(\frac{k}{c}\right) + F_k\left(\frac{k}{c}\right) \right\} \\ &\quad + \sigma^4 \left(c^2 \frac{k+2}{k} - 2c + 1 \right). \end{aligned}$$

Its derivative with respect to c is

$$\frac{\partial L}{\partial c} = 2\sigma^4 \left[(R-1) \left\{ c \frac{k+2}{k} F_{k+4}\left(\frac{k}{c}\right) - F_{k+2}\left(\frac{k}{c}\right) \right\} + c \frac{k+2}{k} - 1 \right],$$

and further differentiation yields

$$\frac{\partial^2 L}{\partial c^2} = 2\sigma^4 \frac{k+2}{k} \left\{ 1 + (R-1) F_{k+4}\left(\frac{k}{c}\right) \right\}.$$

The optimal factor c_R^* is found by the Newton-Raphson algorithm. For $R = 1$, we have $c_1^*(k) = k/(k+2)$, confirming the result of Markowitz (1968) that $k\hat{\sigma}^2/(k+2)$ is the estimator efficient with quadratic loss.

For estimating σ with minimum expected piecewise quadratic loss, the expected loss of $d\hat{\sigma}$ is

$$\begin{aligned} L &= \sigma^2(R-1) \left\{ d^2 F_{k+2}\left(\frac{k}{d^2}\right) - 2 \frac{d\gamma_k}{\sqrt{k}} F_{k+1}\left(\frac{k}{d^2}\right) + F_k\left(\frac{k}{d^2}\right) \right\} \\ &\quad + \sigma^2 \left(d^2 - 2 \frac{d\gamma_k}{\sqrt{k}} + 1 \right). \end{aligned}$$

Its first- and second-order derivatives are

$$\begin{aligned} \frac{\partial L}{\partial d} &= 2\sigma^2 \left[(R-1) \left\{ d F_{k+2}\left(\frac{k}{d^2}\right) - \frac{\gamma_k}{\sqrt{k}} F_{k+1}\left(\frac{k}{d^2}\right) \right\} + d - \frac{\gamma_k}{\sqrt{k}} \right] \\ \frac{\partial^2 L}{\partial d^2} &= 2\sigma^2 \left\{ (R-1) F_{k+2}\left(\frac{k}{d^2}\right) + 1 \right\} \end{aligned}$$

respectively. For $R = 1$, we obtain the factor $d_R^* = \gamma_k/\sqrt{k}$, in agreement with Markowitz (1968).

The functions c_R^* and d_R^* are plotted in Figure 2. The Newton-Raphson algorithm for their evaluation requires 5–7 iterations to achieve precision to eight decimal places. As in Figure 1, the functions c_R^* and d_R^* converge very slowly to unity as $k \rightarrow +\infty$. For $R = 1$, the functions c_1^* and d_1^* are smaller than unity, but so is also c_2^* for $k < 26$, and even $c_5^*(5)$ is very close to unity. This is a consequence of the asymmetry of the χ^2 distribution. Whilst the loss for understatement is bounded by $R\sigma^4$, the loss for overstatement has no upper bound. For highly dispersed $\hat{\sigma}^2$, that is, for large

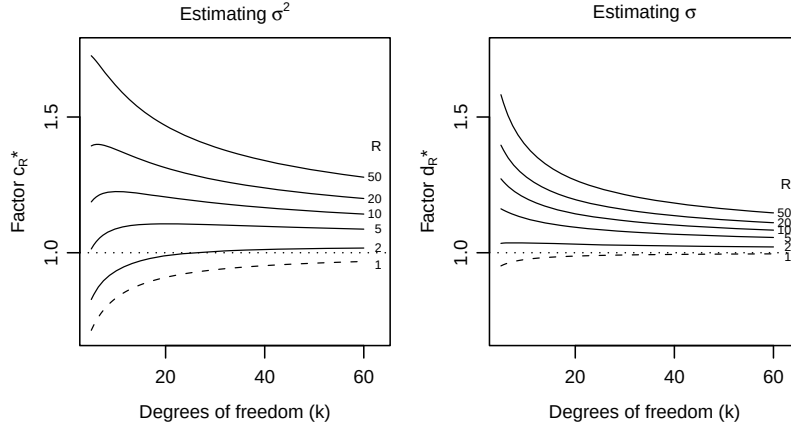


Figure 2: The factors c_R^* and d_R^* for which $c_R^* \hat{\sigma}^2$ and $d_R^* \hat{\sigma}$ are associated with minimum expected piecewise quadratic loss, as functions of the degrees of freedom $k \in [5, 60]$ and aversity R indicated at the right-hand margins.

σ^2 and small k , sizeable contributions to the expected loss are made by $\hat{\sigma}^2 \gg \sigma^2$. That is why the expected loss is an increasing function of k for small k and R up to about 20. For piecewise linear loss, c_R^* is increasing at $k = 5$ only for $R < 4$; $c_2^*(k)$ is close to a constant for $5 < k < 60$.

For estimating the reciprocal of the variance, $1/\sigma^2$, by $1/(c\hat{\sigma}^2)$, a practical choice is the piecewise reciprocal loss function given as $R\{1/(c\hat{\sigma}^2) - 1/\sigma^2\}$ when $c\hat{\sigma}^2 < \sigma^2$ and $1/\sigma^2 - 1/(c\hat{\sigma}^2)$ otherwise. This loss function can equivalently be considered also for σ^2 . For $k > 2$, the expected loss is

$$\begin{aligned} L &= R \int_0^{\sigma^2} \left(\frac{1}{x} - \frac{1}{\sigma^2} \right) \frac{k}{c\sigma^2} f_k \left(\frac{kx}{c\sigma^2} \right) dx + \int_{\sigma^2}^{+\infty} \left(\frac{1}{\sigma^2} - \frac{1}{x} \right) \frac{k}{c\sigma^2} f_k \left(\frac{kx}{c\sigma^2} \right) dx \\ &= \frac{1}{\sigma^2} \left\{ R \int_0^{k/c} \left(\frac{k}{cy} - 1 \right) f_k(y) dy + \int_{k/c}^{+\infty} \left(1 - \frac{k}{cy} \right) f_k(y) dy \right\} \\ &= \frac{1}{\sigma^2} \left[(R+1) \left\{ \frac{1}{c} \frac{k}{k-2} F_{k-2} \left(\frac{k}{c} \right) - F_k \left(\frac{k}{c} \right) \right\} + 1 - \frac{1}{c} \frac{k}{k-2} \right], \end{aligned}$$

an expression similar to (2). Its derivative with respect to $1/c$ is

$$\frac{\partial L}{\partial (1/c)} = \frac{1}{\sigma^2} \frac{k}{k-2} \left\{ (R+1) F_{k-2} \left(\frac{k}{c} \right) - 1 \right\},$$

and so the optimal factor is

$$c_R^* = \frac{k}{F_{k-2}^{-1} \left(\frac{1}{R+1} \right)}.$$

It differs from its counterpart for piecewise linear loss, (3), only by the degrees of freedom, $k-2$ vs. $k+2$. For a given value of R , the piecewise reciprocal loss yields a higher optimal factor c_R^* than the piecewise linear loss, and the difference is substantial for small k . For example, $c_{20}^*(10) = 1.94$

for piecewise linear, and 3.72 for piecewise reciprocal loss. On reflection, such a large difference comes as no surprise, because the loss when $\hat{\sigma}^2 < \sigma^2$ is limited with piecewise linear loss, but not with piecewise reciprocal. Conversely, piecewise linear loss is not limited when $\hat{\sigma}^2 > \sigma^2$, whereas piecewise reciprocal is.

With the piecewise reciprocal loss for estimating σ by $d\hat{\sigma}$, we obtain, by using similar operations, the optimal coefficient

$$d_R^* = \frac{\sqrt{k}}{\sqrt{F_{k-1}^{-1}\left(\frac{1}{R+1}\right)}}.$$

Thus, the choice of the piecewise linear or reciprocal loss function, applied to estimating σ^2 or σ , is equivalent to the choice of the degrees of freedom (gained or lost) in the quantile of the χ^2 distribution.

The errors of estimation can also be considered on a multiplicative scale, regarding the relative error $\hat{\sigma}^2/\sigma^2 - 1$ when $\hat{\sigma}^2 > \sigma^2$ on par with $R(\sigma^2/\hat{\sigma}^2 - 1)$ when $\hat{\sigma}^2 < \sigma^2$, and similarly for σ estimated by $\hat{\sigma}$. The expected loss in estimating σ^2 by $c\hat{\sigma}^2$ is

$$\begin{aligned} L &= R \int_0^{\sigma^2} \left(\frac{\sigma^2}{x} - 1 \right) \frac{k}{c\sigma^2} f_k \left(\frac{kx}{c\sigma^2} \right) dx + \int_{\sigma^2}^{+\infty} \left(\frac{x}{\sigma^2} - 1 \right) \frac{k}{c\sigma^2} f_k \left(\frac{kx}{c\sigma^2} \right) dx \\ &= \frac{R}{c} \frac{k}{k-2} F_{k-2} \left(\frac{k}{c} \right) - (R-1) F_k \left(\frac{k}{c} \right) - c F_{k+2} \left(\frac{k}{c} \right) + c - 1 \end{aligned} \quad (4)$$

for $k > 2$. The first- and second-order derivatives are

$$\begin{aligned} \frac{\partial L}{\partial c} &= 1 - F_{k+2} \left(\frac{k}{c} \right) - \frac{R}{c^2} \frac{k}{k-2} F_{k-2} \left(\frac{k}{c} \right) \\ \frac{\partial^2 L}{\partial c^2} &= \frac{k}{c^3} \left\{ \frac{2R}{k-2} F_{k-2} \left(\frac{k}{c} \right) + (R+1) f_k \left(\frac{k}{c} \right) \right\}. \end{aligned}$$

As an alternative, $\pm \log(\sigma^2/x)$ could be integrated in (4), but numerical integration would then be necessary, in every iteration of the Newton-Raphson algorithm. The expressions for the derivatives are also more complex.

For estimating σ by $d\hat{\sigma}$, we have the following expressions for the expected loss and its derivatives:

$$\begin{aligned} L &= \frac{R\sqrt{k}}{d\gamma_{k-1}} F_{k-1} \left(\frac{k}{d^2} \right) - (R-1) F_k \left(\frac{k}{d^2} \right) - \frac{d\gamma_k}{\sqrt{k}} F_{k+1} \left(\frac{k}{d^2} \right) + \frac{d\gamma_k}{\sqrt{k}} - 1 \\ \frac{\partial L}{\partial d} &= \frac{\gamma_k}{\sqrt{k}} - \frac{R\sqrt{k}}{d^2\gamma_{k-1}} F_{k-1} \left(\frac{k}{d^2} \right) - \frac{\gamma_k}{\sqrt{k}} F_{k+1} \left(\frac{k}{d^2} \right) \\ \frac{\partial^2 L}{\partial d^2} &= \frac{2\sqrt{k}}{d^4} \left\{ \frac{dR}{\gamma_{k-1}} F_{k-1} \left(\frac{k}{d^2} \right) + \sqrt{k}(R+1) f_k \left(\frac{k}{d^2} \right) \right\}. \end{aligned}$$

The functions c_R^* and d_R^* are plotted in Figure 3. Since the relative error $c = \hat{\sigma}^2/\sigma^2$ coincides with $d = \hat{\sigma}/\sigma$ for $d = \sqrt{c}$, one might expect that $d_R^* \doteq \sqrt{c_R^*}$. In the right-hand panel, $\sqrt{c_R^*}$ is plotted by gray colour. The differences $d_R^* - \sqrt{c_R^*}$ are positive throughout, decrease with k and R and are imperceptible for $R = 1$; for example, $d_1^*(5) - \sqrt{c_1^*(5)} = 0.005$.

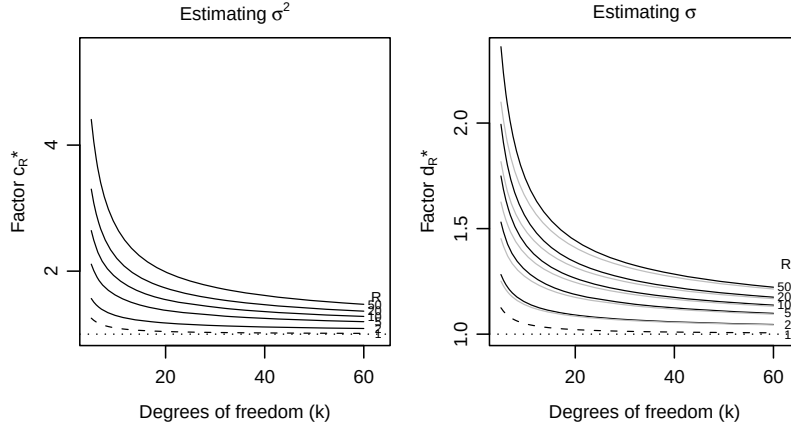


Figure 3: The factors c_R^* and d_R^* for which $c_R^* \hat{\sigma}^2$ and $d_R^* \hat{\sigma}$ are associated with minimum expected loss on the multiplicative scale, as functions of the degrees of freedom $k \in [5, 60]$ and aversity R indicated at the right-hand margins. The factors $\sqrt{c_R^*}$ are drawn in the right-hand panel by gray colour.

3. LINEX loss

The LINEX loss function is defined as

$$l_a(\Delta) = \exp(a\Delta) - a\Delta - 1,$$

where Δ is the error of estimation and the parameter a controls the asymmetry of the loss. For $a < 0$, $l_a(\Delta)$ is close to $\exp(a\Delta)$ when $\Delta \ll 0$ and close to $-a\Delta$ when $\Delta \gg 0$. Zellner (1986) derived estimators of the regression parameters in ordinary regression that have minimum expected loss with respect to LINEX. For applications related to our context, see Chaturvedi, Bhatti and Kumar (2000) and Zou, Zeng, Wan and Guan (2009).

For estimating σ^2 with aversity to understatement, only negative values of a are of interest. We can apply LINEX with $\Delta = \hat{\sigma}^2 - \sigma^2$ or $\Delta = \hat{\sigma}^2/\sigma^2 - 1$. The distribution of the latter does not depend on σ^2 , sparing us some analytical difficulties. The expected loss of $c\hat{\sigma}^2$ with LINEX using $\Delta = \hat{\sigma}^2/\sigma^2 - 1$ is

$$\begin{aligned} L &= \int_0^{+\infty} \{\exp(ax - a) - a(x - 1) - 1\} \frac{1}{\Gamma(\frac{k}{2})} \frac{k}{2c} \left(\frac{kx}{2c}\right)^{k/2-1} \exp\left(-\frac{kx}{2c}\right) dx \\ &= \exp(-a) \left(\frac{k}{k-2ac}\right)^{k/2} - ac + a - 1, \end{aligned}$$

so long as $c < k/(2a)$. The unique root of its derivative,

$$\frac{\partial L}{\partial c} = a \left\{ \exp(-a) \left(\frac{k}{k-2ac}\right)^{k/2+1} - 1 \right\},$$

is

$$c_a^* = \frac{k}{2a} \left\{ 1 - \exp\left(-\frac{2a}{k+2}\right) \right\}, \quad (5)$$

and the minimum of L attained at this point is

$$a - \left(\frac{k}{2} + 1\right) \left\{ 1 - \exp\left(-\frac{2a}{k+2}\right) \right\}.$$

For the expected loss in estimating σ by $d\hat{\sigma}$,

$$\int_0^{+\infty} \left[\exp\left\{ a \left(\frac{d\sqrt{y}}{\sqrt{k}} - 1 \right) \right\} - a \left(\frac{d\sqrt{y}}{\sqrt{k}} - 1 \right) \right] f_k(y) dy,$$

we do not have a closed-form expression. By expanding the exponential in the integrand,

$$\exp(u\sqrt{y}) = \sum_{i=0}^{\infty} \frac{u^i}{i!} y^{i/2},$$

and exchanging the order of integration and summation, we obtain the expression

$$L = \exp(-a) \sum_{i=0}^{\infty} \frac{2^{i/2}}{i!} \left(\frac{ad}{\sqrt{k}} \right)^i \frac{\Gamma\left\{\frac{1}{2}(k+i)\right\}}{\Gamma\left(\frac{1}{2}k\right)} - ad \frac{\sqrt{2}}{\sqrt{k}} \frac{\Gamma\left\{\frac{1}{2}(k+1)\right\}}{\Gamma\left(\frac{1}{2}k\right)} + a - 1, \quad (6)$$

in which the summation converges fast owing to the factor $1/i!$. The first- and second-order derivatives of L have forms similar to (6), because each term in the summation is a scalar multiple of $d^i/i!$. The terms can be evaluated recursively. In the Newton-Raphson algorithm, up to 25 terms of the summation have to be evaluated in each iteration, but 5–7 iterations suffice for $5 \leq k \leq 60$ and $0.1 \leq -a \leq 4.0$.

The optimal factors c_a^* and d_a^* for respective estimation of σ^2 and σ are plotted in Figure 4. Although $c = \hat{\sigma}^2/\sigma^2$ corresponds to d^2 , the functions $c_a^*(k)$ and $d_a^{*2}(k)$ (drawn in the left-hand panel by gray colour) differ appreciably, especially for small k and small values of $-a$. With $a < 0$, the near-exponential loss for negative values of Δ does not represent a harsh penalty because Δ attains large (positive) values with nontrivial probability, especially for small k , whereas $P(\Delta > -1) = 1$. That is why d_a^* exceeds unity only for $a < -2$, and $c_a^* > 1$ only for $a < -1$.

For estimating with LINEX loss and additive error, $\Delta = \hat{\sigma}^2 - \hat{\sigma}^2$ or $\Delta = \hat{\sigma} - \hat{\sigma}$, we obtain expressions similar to (5) and (6), but they involve σ^2 . For example, the expected loss in estimating σ^2 is minimized for

$$c_a^* = \frac{k}{2a\sigma^2} \left\{ 1 - \exp\left(-\frac{2a\sigma^2}{k+2}\right) \right\},$$

so long as $c < k/(2a\sigma^2)$. Substituting $\hat{\sigma}^2$ for σ^2 resolves this problem, but the resulting estimator is problematic for small k . The variance σ^2 is a factor in the upper bound for the value of c , although c^* depends on σ^2 very weakly for small values of $-a\sigma^2$; the Taylor expansion of c_a^* implies that $c_a^* \doteq k/(k+2)$.

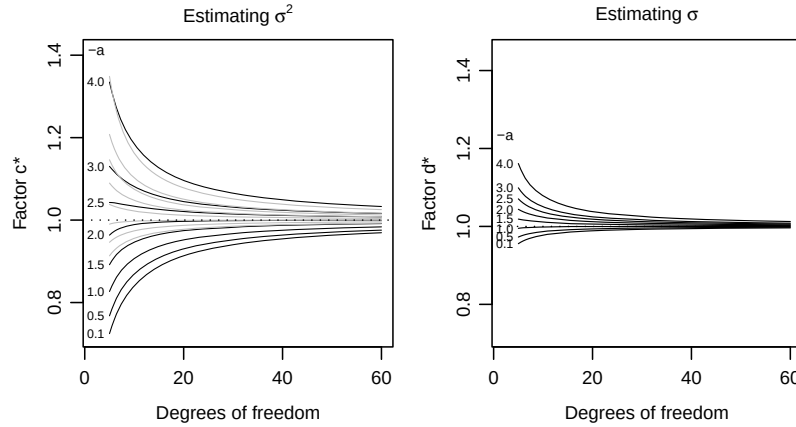


Figure 4: The factors c_R^* and d_R^* for which $c_R^* \hat{\sigma}^2$ and $d_R^* \hat{\sigma}$ are associated with minimum expected LINEX loss with multiplicative error ($\Delta = \hat{\sigma}^2 / \sigma^2 - 1$), as functions of the degrees of freedom $k \in [5, 60]$ and coefficient a indicated at the left-hand margins. The coefficients d_a^{*2} are plotted in the left-hand panel by gray colour.

4. Discussion

Assessment of the precision is an important accompaniment of an estimate. In our perspective, not underestimation (negative bias) but understatement (negative estimation error) is regarded as generally undesirable for sampling variances and standard errors. Since understatement cannot be ruled out, we characterize our or our client's aversity to it by an asymmetric loss function which penalizes understatement more harshly than overstatement, with an aversity coefficient $R > 1$ or the asymmetry coefficient a in the LINEX loss function. The estimators, derived in the previous sections, are biased for the sampling variance or standard error, but they represent a well defined balance of the positive and negative errors of estimation. By construction, all the estimators have the form $b\hat{\sigma}^2$, where the scalar b depends only on the degrees of freedom k and the coefficient R or a , except for one version of the LINEX loss function.

For piecewise linear, quadratic, reciprocal and multiplicative loss, the factors $b_R(k)$ converge very slowly to 1.0 as $k \rightarrow +\infty$, and differ from it appreciably even for $k = 60$ (Figures 1–3). The factors attain very large values with large R and small k , when the substantial dispersion of $\hat{\sigma}^2$ renders the property of no bias of next to no value in any reasonable perspective.

The estimators we derived have minimum expected loss within the classes of estimators that are scalar multiples of $\hat{\sigma}^2$ and $\hat{\sigma}$. There may be estimators of σ^2 or σ with expected loss smaller than our proposals, but they are bound to be much more complex functions of $\hat{\sigma}^2$, and we have no means of identifying them.

The piecewise linear, quadratic and reciprocal loss functions are constructed for the specific purpose of estimating with aversity to understatement (or overstatement), but they are not smooth at zero. In contrast, LINEX loss functions are smooth throughout, but their appeal for estimation with aversity is less direct because there is no simple expression for the difference $l_a(\Delta) - l_a(-\Delta)$ or

ratio $l_a(\Delta)/l_a(-\Delta)$. For $\Delta > 0$, the difference has the same sign as a , but neither the difference nor the ratio has a simple description that might be used with effect in the process of eliciting a suitable value of the coefficient a .

Our derivations are based on the assumption that $\hat{\sigma}^2$ has a scaled χ^2 distribution, as in ordinary regression and its special cases (e.g., comparison of two normal random samples). Estimators of the sampling variances in some other models, such as generalized linear and mixed models, have χ^2 distributions only approximately (McCullagh and Nelder, 1989; Longford, 2000), so our results do not apply to them with precision.

In a practical setting, the analyst and client have to agree on a loss function (the type, such as piecewise linear, and the aversity R). Greater integrity may be achieved if they settle only on the type and a range of aversity coefficients, which lead to a range of estimates $b\hat{\sigma}^2$. See (Longford, 2010; Longford, 2012) for examples of using such sets of plausible loss functions. Reasonable choices for the aversity coefficient R are several times greater than unity. Estimators close to the conventional estimators $\hat{\sigma}$ and $\hat{\sigma}^2$ are obtained for $R = 1$. The fact that this is an extreme choice, in discord with the aversity to understatement, suggests that we should review the practice of (approximately) unbiased estimation of σ and σ^2 .

References

- BERGER, J. O. (1985). *Statistical Decision Theory and Bayesian Analysis*. 2nd edition. Springer-Verlag, Berlin.
- CHATURVEDI, A., BHATTI, M. I., AND KUMAR, K. (2000). Bayesian analysis of disturbances variance in the linear regression model under asymmetric loss functions. *Applied Mathematics and Computation*, **114**, 149–153.
- DEGROOT, M. H. (1970). *Optimal Statistical Decisions*. 2nd edition. McGraw-Hill, New York.
- LINDLEY, D. V. (1985). *Making Decisions*. Wiley and Sons, Chichester, UK.
- LONGFORD, N. T. (2000). On estimating standard errors in multilevel analysis. *Journal of the Royal Statistical Society Series D*, **49**, 389–398.
- LONGFORD, N. T. (2010). Bayesian decision making about small binomial rates with uncertainty about the prior. *The American Statistician*, **64**, 164–169.
- LONGFORD, N. T. (2012). Comparing normal random samples, with uncertainty about the priors and utilities. *Scandinavian Journal of Statistics*, **39**, 729–742.
- MARKOWITZ, E. (1968). Minimum mean-square-error of estimation of the standard deviation of the normal distribution. *The American Statistician*, **22**, 26.
- MCCULLAGH, P. AND NELDER, J. A. (1989). *Generalized Linear Models*. 2nd edition. Chapman and Hall, London.
- VARIAN, H. R. (1975). A Bayesian approach to real estate assessment. In FIENBERG, S. E. AND ZELLNER, A. (Editors) *Studies in Bayesian Econometrics and Statistics in Honour of Leonard J. Savage*. North-Holland, Amsterdam, pp. 195–208.
- ZELLNER, A. (1986). Bayesian estimation and prediction using asymmetric loss functions. *Journal of the American Statistical Association*, **81**, 446–451.

ZOU, G., ZENG, J., WAN, A., AND GUAN, Z. (2009). Stein-type improved estimation of standard error under asymmetric linex loss function. *Statistics*, **43**, 121–129.

